Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	**TE-AI&DS**	**Semester:**	**V**
Course Code:	**2015118**	**Course Name:**	**Artificial Intelligence & Soft Computing Lab**

Name of Student:	**Shreyas Sawant**
Roll No. :	**78**
Experiment No.:	**1**
Title of the Experiment:	**Design PEAS Framework using relevant AI Problem Statement**
Date of Performance:	**13/07/26**
Date of Submission:	**27/07/26**

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : Dr. Vaishali Shirsath

Signature :

Date :

Aim: To select a real-world problem relevant to Artificial Intelligence, identify its characteristics, describe it using the PEAS (Performance Measure, Environment, Actuators and Sensors) framework and perform AI problem formulation.

Objectives:

After completing this experiment, the student will be able to:

1. Understand the concept of Artificial Intelligence and Intelligent Agents.
2. Identify real-world problems that can be solved using AI.
3. Analyze an AI problem using the PEAS framework.
4. Perform problem formulation for an intelligent agent.
5. Develop logical thinking for AI-based system design.

Tools Required:

- **Hardware:**
  - Desktop/Laptop
  - Internet Connection
- **Software:**
  - Microsoft Word
  - Draw.io / Lucidchart (Optional)
  - VS Code (Optional)

Theory:

Artificial Intelligence (AI) is a branch of computer science that focuses on developing systems capable of performing tasks that normally require human intelligence, such as reasoning, learning, problem-solving, perception and decision-making. An Intelligent Agent is a system that interacts with its environment by perceiving information through sensors and taking appropriate actions using actuators to achieve a specific goal.

Before implementing an AI solution, it is essential to formulate the problem correctly. One of the standard methods used for this purpose is the PEAS (Performance Measure, Environment, Actuators and Sensors) framework. The PEAS framework helps define the operating environment of an intelligent agent and specifies how the agent should interact with its surroundings. It provides a systematic approach to identify the objectives of the agent, the environment in which it operates, the actions it can perform, and the information it requires for decision-making.

Proper problem formulation using the PEAS model forms the foundation for designing efficient and intelligent AI systems and is widely used in applications such as autonomous vehicles, healthcare systems, robotics, smart homes and virtual assistants.

Procedure:

1. Study the fundamentals of Artificial Intelligence and Intelligent Agents.
2. Select a real-world problem that can be solved using Artificial Intelligence.
3. Clearly define the selected problem and identify the intelligent agent.
4. Analyze the problem using the PEAS framework by identifying the Performance Measure, Environment, Actuators and Sensors.
5. Perform AI problem formulation by defining the initial state, goal state, possible actions and constraints.
6. Prepare the PEAS table and document the problem formulation.
7. Verify the completeness of the analysis and record the observations.

Description of the Experiment:

AI Problem Statement : AI-Powered ICU Patient Monitor

The **AI-Powered ICU Patient Monitor** is a clinical software agent designed to track patient health in high-stakes, real time hospital settings. Instead of just sounding an alarm after a patient's vitals crash, this intelligent system continuously fuses streams of physiological data (like ECG waveforms, blood pressure, oxygen levels, and lab results) to predict life-threatening clinical events such as sepsis, respiratory failure, or cardiac arrest hours before they manifest outwardly. By transforming reactive medical equipment into a proactive predictive tool, it gives doctors and nurses a critical window of time to intervene, dramatically reducing hospital mortality rates while filtering out baseline noise to minimize clinical alarm fatigue.

Detailed Description:

PEAS Framework Analysis:

PEAS Component	Detailed Operational Attributes
Performance Measure	**Prediction Accuracy:** Maximize sensitivity/recall for life-threatening conditions (e.g., Sepsis, Cardiac Arrest). **False Alarm Minimization:** Lower the false positive rate to prevent clinical "alarm fatigue." **Early Detection Window:** Maximize lead time (e.g., predict systemic deterioration 4–6 hours before onset). **Data Uptime:** Maintain 100% processing availability without system latency.

Environment	**Physical Setting:** The Intensive Care Unit (ICU) clinical ward and patient bed areas. **Target System:** The patient's dynamic internal biological/physiological state. **Human Elements:** On-duty intensive care physicians, nurses, and rapid response medical teams. **Data Infrastructure:** Hospital Electronic Health Records (EHR) networks and lab information systems.
Actuators	**Visual Displays:** Real-time dashboards showing patient stability index trends and risk percentages. **Alert Subsystems:** High-priority auditory alarms and flashing color-coded UI indicators at nursing stations. **Communication Triggers:** Automated high-priority text alerts and telemetry routing to on-call physician pagers. **Electronic Records:** Automated diagnostic entry logging within the patient's centralized EHR file.
Sensors	**Continuous Physiological Feeds:** Electrocardiogram (ECG/EKG) tracking, Photoplethysmogram (SpO_2), Arterial Line Blood Pressure monitors, and core thermal probes. **Discrete Electronic Data Inputs:** Real-time text/numeric feeds of incoming chemistry lab results (e.g., lactate, white blood cell count). **Manual Input Streams:** Nurse-logged flowsheets monitoring fluid intake/output metrics.

AI Problem Formulation:

To frame this system as a computational search and optimization problem, the system architecture is formulated through these strict parameters:

- **Initial State:** The agent establishes a connection to a newly admitted patient profile. The current vital stream matrix reads steady baseline physiological values, historical medical records are parsed, and the real-time "Patient Risk Index" is initialized at baseline (0%).
- **Goal State:** The agent continuously executes its analysis loop throughout the patient's entire ICU stay, maintaining an anomaly detection accuracy threshold that successfully alerts medical staff of all critical health declines well ahead of outward physical crashes, while registering zero system downtime.
- **Possible Actions (Transition Operators):**
    - AnalyzeSensorStream(): Compute moving averages, cross-correlate vitals, and update the predictive health trajectory model.
    - SuppressAlarm(): Maintain ambient logging if vital anomalies are determined to be

non-critical artifacts (e.g., patient movement shifting an ECG lead).
    - TriggerLowPriorityAlert(): Flag early, slow-moving deterioration trends on the central dashboard for routine nurse assessment.
    - EscalateToCriticalAlarm(): Broadcast immediate audible and pager alerts if the predictive probability of an event crosses the emergency threshold.
- **Constraints:**
    - **Latency Limit:** The processing time delta between a sensor reading an anomalous vital signature and the actuator firing an alert must be less than 1 second.
    - **Regulatory Compliance:** Absolute data encryption in transit and at rest to comply with strict medical privacy laws (e.g., HIPAA).
    - **Clinical Safety Interlock:** The AI software cannot directly alter life-support machinery parameters (e.g., mechanical ventilator volume) without explicit manual physician verification.

Verification of Completeness & Recorded Observations:

System Verification:

The analysis confirms that all primary inputs, operational bounds, and processing steps required to implement this software agent are clearly accounted for. The agent safely maps physiological inputs directly to non-invasive clinical alerting actions.

Recorded Observations:

1. **High Environmental Stochasticity:** A patient's physiological state is highly dynamic and unpredictable. Subtle variances in input streams (like a sudden drop in SpO_2 combined with an elevated heart rate) can signify completely different clinical outcomes depending on the underlying EHR history, making context-aware deep learning models essential.
2. **The Alarm Fatigue Trade-off:** The agent faces a constant optimization paradox within its path-cost logic: setting alert thresholds too low results in hundreds of daily false alarms that cause human staff to ignore the system, while setting thresholds too high risks missing a fatal event. The performance tuning must emphasize precise balance through robust multi-parameter anomaly cross-verification.

Conclusion:

Problem formulation for the selected **AI-Powered ICU Patient Monitor** application was successfully performed using the PEAS framework. The analysis provided a highly structured and rigorous understanding of the intelligent software agent, its complex physiological and clinical operating environment, and the critical sensory and alerting components required to design a predictive healthcare solution. By defining the agent's initial states, goal spaces, possible transition actions, and strict medical constraints, this experiment successfully establishes the engineering and algorithmic foundation necessary before implementing deep learning or real-time anomaly detection models in a critical care setting.

Questions to be Answered by Students:

1. Which real-world problem was selected for Artificial Intelligence problem formulation?

Ans- The selected real-world problem is **AI Powered ICU Patient Monitor**. The system uses Artificial Intelligence to continuously monitor ICU patients' vital signs, detect abnormalities, predict health deterioration, and alert medical staff in real time, improving patient safety and reducing response time.

2. What are the four components of the PEAS framework identified for the selected problem?

Ans-The **PEAS (Performance Measure, Environment, Actuators, Sensors)** framework for an AI Powered ICU Patient Monitor is:

Component	Description
Performance Measure (P)	Accurate patient monitoring, early detection of critical conditions, minimal false alarms, quick emergency response, improved patient outcomes.
Environment (E)	ICU rooms, patients, doctors, nurses, medical equipment, hospital network, electronic health records (EHR).
Actuators (A)	Alarm system, notifications to doctors/nurses, dashboard updates, automated emergency alerts, medication recommendations (if supported).
Sensors (S)	ECG monitors, heart rate sensors, blood pressure monitors, pulse oximeter (SpO_2), temperature sensors, respiratory rate sensors, wearable devices.

3. What is the goal state of the selected AI problem?

Ans-The goal state is:

T**o** continuously monitor ICU patients, accurately detect and predict critical health conditions in real time, and immediately alert healthcare professionals to enable timely medical intervention, thereby improving patient safety and survival rates.

4. State two observations obtained from the PEAS analysis of the selected problem.

Ans-

1. The system operates in a dynamic and real-time environment, where patient health conditions can change rapidly, requiring continuous monitoring and immediate decision-making.
2. Multiple sensors provide continuous patient data, enabling the AI system to analyze vital signs, detect abnormalities early, and generate timely alerts for doctors and nurses.